Diego De Pablo

Bioinformatics Engineer | Software, Infrastructure & Data Science

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Professional Summary

Bioinformatics graduate and Master's student in Computer Engineering. Specialized in backend engineering, data architectures, distributed systems, and applied AI, with experience in real-time telemetry and edge devices. Focused on delivering technical rigor, scalability, and continuous learning to high-impact projects.

Professional Experience

Software Engineer, Khaos Research
Sept 2025 – Present

R&D on European federated data spaces (SEDIA | Gaia-X)

- Designed and integrated a hybrid search engine combining vector retrieval in **Qdrant** and lexical search, supported by knowledge-graph agents to navigate data spaces.
- Proposed and developed a metadata extraction and cataloging tool, evolving into an independent project with an ongoing scientific publication.
- Optimized Sentinel-2 satellite imagery ingestion by 80% (LandCoverPy) designing a distributed divide-and-conquer pipeline (tile/season sharding) orchestrated with Dask.

Full-Stack Developer, mSurgery
March 2025 – July 2025

Medical telemetry platforms and real-time surgical digital twins

- Led the frontend migration to **Svelte 5**, adopting the reactive runes paradigm.
- Delivered technical demonstrations and presentations in English to international investors and industry committees at the National Digital Content Hub.

Junior Software Engineer, T&B Company Group
June 2021 – December 2021

Corporate platform modernization toward REST microservices

- Refactored legacy modules in Java and Angular toward decoupled architectures.
- Designed standard interface contracts with **OpenAPI / Swagger** and documented REST microservices to accelerate frontend-backend team integration.

Selected Technical Projects

diegodepablo.is-a.dev/projects

Biomedical Telemonitoring IoT Wristband (B.Sc. Thesis)
Stack: C++, FreeRTOS, ESP32, MQTT, FastAPI, TimescaleDB, Docker

- End-to-end telemonitoring prototype: multithreaded ESP32 firmware for vital signs via photoplethysmography (pulse and SpO2) and fall detection algorithm using MPU-6050 accelerometer.
- Wireless synchronization pipeline over MQTT to a FastAPI backend and TimescaleDB with a real-time web dashboard. **Grade:** 10/10 with Honors.

TITAN Workflow: Scientific Workflow Orchestrator & Assistant
Stack: FastAPI, Pydantic AI, LangChain, LLMs, Docker

- Intelligent assistant for scientific workflow composition and validation: verifies data schemas and dependency compatibility prior to execution using Pydantic AI validators.

Education

M.Sc. in Computer Engineering | *University of Malaga* — **2026 – Present**

B.Sc. in Health Engineering (Bioinformatics Specialization) | *University of Malaga* — **Sep 2021 – Jul 2025**

GPA: **7.96/10**. Honors in *Programming Fundamentals*, *Intelligent Systems*, and *Bioinformatics Projects*. Thesis graded **10/10 with Honors**.

Technical Skills & Languages

Programming Languages: Python, JavaScript, TypeScript, C#, SQL, Java, Bash, R.

Backend & Architecture: FastAPI, REST Microservices, OpenAPI / Swagger, Pydantic, Concurrency (AsyncIO), Node.js.

Databases & Search: PostgreSQL, Elasticsearch, Qdrant (Vector DB), Redis.

AI, DevOps & Systems: Pydantic AI, LangChain, RAG / Embeddings, LLM Agents, PyTorch, Docker, Kubernetes, Helm, Argo CD, CI/CD, Linux.

Frontend & UI: React, Svelte 5, Next.js, HTML5, CSS, Tailwind CSS.

Languages: Spanish (Native) · **English** (B2 Professional).